

$$l = \lim_{p \rightarrow \infty} \frac{(2^p p!)^4 \times 2}{\pi (2p+1)! (2p)!} \quad \left[\text{Théorème de Wallis} \right]$$

$$= \lim_{p \rightarrow \infty} \frac{2^{4p} (2^p p!)^4 \left(\frac{1}{e}\right)^{4p} \times 2}{\pi \sqrt{2\pi(2p+1)} \left(\frac{2p+1}{e}\right)^{2p+1} \times \sqrt{2\pi \times 2p} \left(\frac{2p}{e}\right)^{2p}}$$

$$= 2 \lim_{p \rightarrow \infty} 2^{4p} \times e^{4p \ln(1/e)} \times e^{-6p \ln(2p+1)} \times e^{-2p \ln(1/e)} \times e^{4p \ln p} \times e^{2p \ln 2}$$

$$\frac{-4p + 2p + 1}{2p}$$

$$= 2 \times \lim_{p \rightarrow \infty} e^{4p \ln p + \ln p + 1 - (2p+1) \ln(2p+1) + 2p \ln 2}$$

$$= 2 \times \lim_{p \rightarrow \infty} e^{-\frac{2}{m} \ln m - \ln m + \ln e - \left(\frac{2}{m} + 1\right) (\ln(2+m) - \ln m) + \frac{2}{m} \ln 2}$$

$$= 2 \times \lim_{m \rightarrow 0^+} e^{\ln e - \left(\frac{2}{m} + 1\right) \ln m + \left(\frac{2}{m} + 1\right) \ln m - \ln(2+m) \left(\frac{2}{m} + 1\right) + \frac{2}{m} \ln 2} \quad \text{où } m = \frac{1}{p}$$

$$= 2 \times \lim_{m \rightarrow 0^+} e^{\frac{2}{m} \ln \frac{2}{2+m} - \ln(2+m)}$$

$$= 2 \times e^{-\ln 2} \times \lim_{m \rightarrow 0^+} e^{-\frac{2}{m} \ln \left(\frac{m}{2} + 1\right)}$$

$$= e \times \lim_{y \rightarrow 0^+} e^{-\frac{\ln(1+y)}{y}}$$

$$= e \times e^{-1}$$

$$l = 1$$

$$W_p W_{p+1} = \frac{\pi}{2} \frac{(2p)!}{(2p+1)!} = \frac{\pi}{2(2p+1)}$$

$$W_n \sim \sqrt{\frac{\pi}{2n}}$$

$$\lim_{n \rightarrow \infty} \frac{W_p W_{p+1}}{2(2p+1)} = 1 \quad \text{d'où}$$

$$\lim_{n \rightarrow \infty} \frac{W_p^2}{2(2p+1)} = 1 \Rightarrow \lim_{n \rightarrow \infty} \frac{W_p}{\sqrt{2(2p+1)}} \times \frac{1}{\sqrt{2p}} = 1$$

$$\lim_{n \rightarrow \infty} \frac{W_n}{\sqrt{\frac{\pi}{2n}}} = 1$$

$$W_{2p} W_{2p+1} = \frac{\pi/2}{(2p+1)} = \frac{\pi/2}{2p} \times \frac{2p}{2p+1} \quad \text{et } W_{2p+1} W_{2p+2} = \frac{\pi/2}{2p+2}$$

$$W_{2p} W_{2p+1} =$$

$$W_{2p}^2 = \frac{\pi/2}{2p} \times \frac{W_{2p}}{W_{2p+1}} \times \frac{2p}{2p+1}$$

$$\frac{W_{2p}}{\sqrt{\frac{\pi}{2(2p)}}} = \frac{W_{2p}}{W_{2p+1}} \times \frac{2p}{2p+1}$$

$$\text{car } \lim_{p \rightarrow \infty} \frac{W_{2p}}{\sqrt{\frac{\pi}{2(2p)}}} = 1 \times 1 = 1$$

$$\text{or } W_{2p} \sim \sqrt{\frac{\pi}{2(2p)}} \quad \text{d'où } W_{2p} \sim \sqrt{\frac{\pi}{2(2p)}}$$

or W_n et W_{n+1} pour n grand

$$\text{d'où } W_n \sim W_{2p} \quad \text{d'où } W_n \sim \sqrt{\frac{\pi}{2n}}$$

$$n! \sim C \sqrt{n} \left(\frac{n}{e}\right)^n$$

$$\text{et } W_{2p} = \frac{\pi}{2} \frac{(2p)!}{(2p+1)!} \sim \frac{\pi}{2} \frac{C \sqrt{2p} (2p/e)^{2p}}{(2^p C \sqrt{p} (p/e)^p)^2} = \frac{\pi}{\sqrt{2} \sqrt{p}} C$$

$$\text{car } C = \lim_{p \rightarrow \infty} \frac{\pi}{\sqrt{2p} W_{2p}} = \lim_{p \rightarrow \infty} \frac{\pi}{\sqrt{2p}} \times \sqrt{\frac{2p}{\pi}} = \sqrt{2\pi}$$

$$\text{Donc } n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

$$e^u = (1+u/n)^n \quad f'(u) = e^u = (1+u/n)^{n-1} \quad e^u = (1+u)$$

$$e^u = e^1$$

$$-n < m < 1$$

$$\frac{1}{1-n} < \frac{1}{1+m} < \frac{1}{2}$$

$$\frac{1}{1-n} < \frac{1}{2}$$

$$\frac{n! e^n}{n^n n}$$

$$\frac{dy}{y} = \frac{dx}{x}$$

$$\ln y = \ln x + C$$

$$y = K e^x$$

$$y^2 = K$$

$$y = \sqrt{K}$$

$$A e^x$$

$$f(n+h) \sim f(n) (1+h/n)$$

$$f(n + \frac{n}{n}) \sim f(n) (1 + \frac{n}{n})$$

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$$f(n) \sim f(n + \frac{n}{n})$$

$$f(\frac{1}{n}) \sim f(n)$$

$$f(n+h) \sim f(n) (1+h/n)$$

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